

3. Gradient Descent

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3.1 The Grid Method

- Let $h : [0, 1]^d \rightarrow \mathbb{R}$ be an L -Lipschitz function; that is,

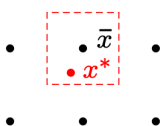
$$|h(x) - h(y)| \leq L \|x - y\|_2 \quad \text{for some constant } L \geq 0$$

- Try to minimize h with the **Grid Method**:

evaluate h at the $(n + 1)^d$ grid points $(i_1/n, \dots, i_d/n)$, $0 \leq i_1, \dots, i_d \leq n$ and solve

$$\min_{0 \leq i_1, \dots, i_d \leq n} h(i_1/n, \dots, i_d/n)$$

- Denote by $\bar{x}$ the grid point closest to the minimizer $x^* \in [0, 1]^d$ of h



Then $\|\bar{x} - x^*\|_2 \leq \frac{\sqrt{d}}{2n}$; Therefore,

$$\min_{0 \leq i_1, \dots, i_d \leq n} h(i_1/n, \dots, i_d/n) \leq h(\bar{x}) \leq h(x^*) + L \frac{\sqrt{d}}{2n}$$

- For an accuracy of $\varepsilon > 0$, one needs $n = \left\lceil \frac{L\sqrt{d}}{2\varepsilon} \right\rceil$, and therefore, $\left(\left\lceil \frac{L\sqrt{d}}{2\varepsilon} \right\rceil + 1 \right)^d$ function evaluations

- This becomes infeasible for high dimensions d (**Curse of Dimensionality**)

3.2 Convex Functions

- A function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is said to be **convex** if

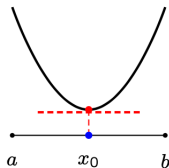
$$h(\lambda x + (1 - \lambda)y) \leq \lambda h(x) + (1 - \lambda)h(y) \quad \text{for all } x, y \in \mathbb{R}^d \text{ and } \lambda \in (0, 1)$$

- Examples** $h(x) = \sum_{i=1}^d |x_i|^p$ for $p \geq 1$ or $h(x) = \|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$ for $p \geq 1$

- Every norm $\|\cdot\|$ is convex. Indeed, $\|\lambda x + (1 - \lambda)y\| \leq \|\lambda x\| + \|(1 - \lambda)y\| = \lambda \|x\| + (1 - \lambda) \|y\|$
- A **subgradient** of h at $x \in \mathbb{R}^d$ is a vector $g \in \mathbb{R}^d$ such that

$$h(x + y) \geq h(x) + \langle y, g \rangle \quad \text{for all } y \in \mathbb{R}^d$$

- The **subdifferential** $\partial h(x)$ is the set of all subgradients of h at x
- If h is convex, $\partial h(x)$ is for all $x \in \mathbb{R}^d$, a **non-empty, convex** subset of $\mathbb{R}^d$
- A convex function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable at $x \Leftrightarrow \partial h(x)$ consists of a single element $\nabla h(x)$
- $x^* \in \mathbb{R}^d$ is a minimizer of a convex function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ if and only if $0 \in \partial h(x^*)$
(0 is a subgradient of h at x^*)



3.3 Gradient Descent

- Let $h : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function with a minimizer $x^* \in \mathbb{R}^d$
- **Start** at $x_0 \in \mathbb{R}^d$ (can be deterministic or random)
- **Gradient steps** $x_{k+1} = x_k - \eta_k \nabla h(x_k)$, $k \geq 0$, for subgradients $\nabla h(x_k)$ and a learning rate $\eta_k > 0$
- Then $h(x^*) - h(x_k) \geq \langle x^* - x_k, \nabla h(x_k) \rangle$, and therefore,
$$2\eta_k (h(x_k) - h(x^*)) \leq 2 \langle x_k - x^*, \eta_k \nabla h(x_k) \rangle = \|x_k - x^*\|_2^2 - \|x_k - x^* - \eta_k \nabla h(x_k)\|_2^2 + \eta_k^2 \|\nabla h(x_k)\|_2^2$$
$$= \|x_k - x^*\|_2^2 - \|x_{k+1} - x^*\|_2^2 + \eta_k^2 \|\nabla h(x_k)\|_2^2$$
- Hence,
$$\sum_{k=0}^K 2\eta_k (h(x_k) - h(x^*)) \leq \|x_0 - x^*\|_2^2 - \|x_{K+1} - x^*\|_2^2 + \sum_{k=0}^K \eta_k^2 \|\nabla h(x_k)\|_2^2$$
- If $\|\nabla h(x_k)\|_2 \leq L$ for all k , then

$$\min_{0 \leq k \leq K} h(x_k) - h(x^*) \leq \frac{\sum_{k=0}^K \eta_k (h(x_k) - h(x^*))}{\sum_{k=0}^K \eta_k} \leq \frac{\|x_0 - x^*\|_2^2 + L^2 \sum_{k=0}^K \eta_k^2}{2 \sum_{k=0}^K \eta_k}$$

- So, if $\sum_{k=0}^{\infty} \eta_k^2 < \infty$ and $\sum_{k=0}^{\infty} \eta_k = \infty$, then $\min_{0 \leq k \leq K} h(x_k) \rightarrow h(x^*)$ for $K \rightarrow \infty$
e.g. $\sum_{k=1}^{\infty} \frac{1}{k^2} < \infty$ and $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$

Finite Computing Budget

- If the number of gradient steps K is fixed, choose η_k to minimize

$$\frac{\|x_0 - x^*\|_2^2 + L^2 \sum_{k=0}^K \eta_k^2}{2 \sum_{k=0}^K \eta_k}$$

- **Exercise:** $\eta_k = \frac{\|x_0 - x^*\|_2}{L\sqrt{K+1}}$ for all $k = 0, 1, \dots, K$ and $\min_{0 \leq k \leq K} h(x_k) - h(x^*) \leq \frac{L \|x_0 - x^*\|_2}{\sqrt{K+1}}$

- So, for accuracy $\varepsilon > 0$, one needs $K = \left\lceil \left(\frac{L \|x_0 - x^*\|_2}{\varepsilon} \right)^2 \right\rceil$ gradient steps

(does not depend on the dimension!)

3.4 Stochastic Gradient Descent

- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space
- Consider a function $H : \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}$, $(\omega, \theta) \mapsto H(\omega, \theta)$ that is
 - convex in θ
 - measurable in ω
 - $\mathbb{E} |H(\theta)| < \infty$ for all $\theta \in \mathbb{R}^d$
- **Exercise** $h(\theta) = \mathbb{E} H(\theta)$ is convex and for every measurable subgradient $\nabla H(\theta)$ ($= \nabla_{\theta} H(\omega, \theta)$),
 $\nabla h(\theta) = \mathbb{E} [\nabla H(\theta)]$ is a subgradient of h at θ
- $\nabla h(\theta)$ might be difficult (impossible) to compute
- Let $H_i, i \geq 1$, be independent copies of H
- Fix $I \in \mathbb{N}$ and set $g_k(\theta) = \frac{1}{I} \sum_{i=kI+1}^{(k+1)I} \nabla H_i(\theta) \approx \nabla h(\theta)$, $k \geq 0$
 - For $I = 1$: *standard SGD*
 - For $I \geq 2$: *SGD with mini-batches* (reduces the variance of the gradient estimates)

- **Start** at $\theta_0 \in \mathbb{R}^d$ (can be deterministic or random independent of $H_1, H_2, \dots$)
- **Stochastic gradient steps** $\theta_{k+1} = \theta_k - \eta_k g_k(\theta_k)$, $k \geq 0$, ($\theta_0, \theta_1, \dots$ are random)
- Denote $\mathcal{F}_k = \sigma(\theta_0, \dots, \theta_k)$. θ_k is $\mathcal{F}_k$ -measurable $\Rightarrow \mathbb{E}[g_k(\theta_k) \mid \mathcal{F}_k] = \nabla h(\theta_k)$
- If h has a minimizer $\theta^* \in \mathbb{R}^d$, then $h(\theta^*) - h(\theta_k) \geq \mathbb{E}[\langle \theta^* - \theta_k, g_k(\theta_k) \rangle \mid \mathcal{F}_k]$, and therefore,

$$\begin{aligned}
 2\eta_k (h(\theta_k) - h(\theta^*)) &\leq \mathbb{E}[2 \langle \theta_k - \theta^*, \eta_k g_k(\theta_k) \rangle \mid \mathcal{F}_k] \\
 &= \|\theta_k - \theta^*\|_2^2 - \mathbb{E}[\|\theta_k - \theta^* - \eta_k g_k(\theta_k)\|_2^2 \mid \mathcal{F}_k] + \eta_k^2 \mathbb{E}[\|g_k(\theta_k)\|_2^2 \mid \mathcal{F}_k] \\
 &= \|\theta_k - \theta^*\|_2^2 - \mathbb{E}[\|\theta_{k+1} - \theta^*\|_2^2 \mid \mathcal{F}_k] + \eta_k^2 \mathbb{E}[\|g_k(\theta_k)\|_2^2 \mid \mathcal{F}_k]
 \end{aligned}$$

- If $\mathbb{E}[\|g_k(\theta_k)\|_2^2 \mid \mathcal{F}_k] \leq L^2$ for all $k = 0, 1, \dots$, then

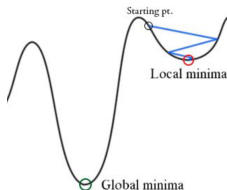
$$\begin{aligned}
 \sum_{k=0}^K 2\eta_k \mathbb{E}[h(\theta_k) - h(\theta^*)] &\leq \mathbb{E}\|\theta_0 - \theta^*\|_2^2 - \mathbb{E}\|\theta_{K+1} - \theta^*\|_2^2 + L^2 \sum_{k=0}^K \eta_k^2 \\
 \Rightarrow \mathbb{E}\left[\min_{0 \leq k \leq K} h(\theta_k)\right] - h(\theta^*) &\leq \frac{\sum_{k=0}^K \eta_k \mathbb{E}[h(\theta_k) - h(\theta^*)]}{\sum_{k=0}^K \eta_k} \leq \frac{\mathbb{E}\|\theta_0 - \theta^*\|_2^2 + L^2 \sum_{k=0}^K \eta_k^2}{2 \sum_{k=0}^K \eta_k}
 \end{aligned}$$

- So, if $\sum_{k=0}^{\infty} \eta_k^2 < \infty$ and $\sum_{k=0}^{\infty} \eta_k = \infty$, then $\mathbb{E}\left[\min_{0 \leq k \leq K} h(\theta_k)\right] \rightarrow h(\theta^*)$ for $K \rightarrow \infty$

Remarks

- One can also show that for convex h , $\min_{0 \leq k \leq K} h(\theta_k) \rightarrow h(\theta^*)$ almost surely for $K \rightarrow \infty$

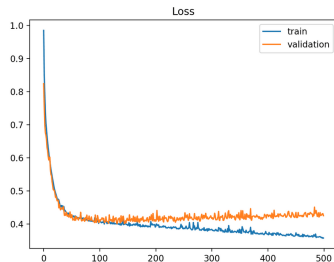
- SGD is also used for non-convex problems;
e.g. to train neural networks ... *even if it is
not guaranteed to converge to a global minimum*



- In practice, various adaptive updating rules are used
- E.g., one can use a schedule of the form
 - $\eta_k = \varepsilon, \quad k = 0, 1, \dots, K_1$
 - $\eta_k = \varepsilon/10, \quad k = K_1 + 1, \dots, K_2$
 - and so on ...
- Or one can start with $\eta_k = \varepsilon$ until the objective does not decrease anymore and then set $\eta_k = \varepsilon/10$...
- Or RMSProp or Adam ... *(see later)*

3.5 Validation

- let $\tilde{H}_i, i = 1, \dots, v$, be independent copies of H independent of $H_i, i \geq 1$ (*validation data set*)
- monitor the *empirical loss* $\frac{1}{v} \sum_{i=1}^v \tilde{H}_i(\theta_k)$ of $\theta_k, k = 0, 1, \dots$ on the *validation data set*
- if the *validation loss* stops decreasing, decrease the learning rate η_k
- if the *validation loss* starts to increase, stop the SGD



- if a *validation data set* is used in training, the resulting θ_k still has to be evaluated on an independent *test data set*